

Study Squad AI

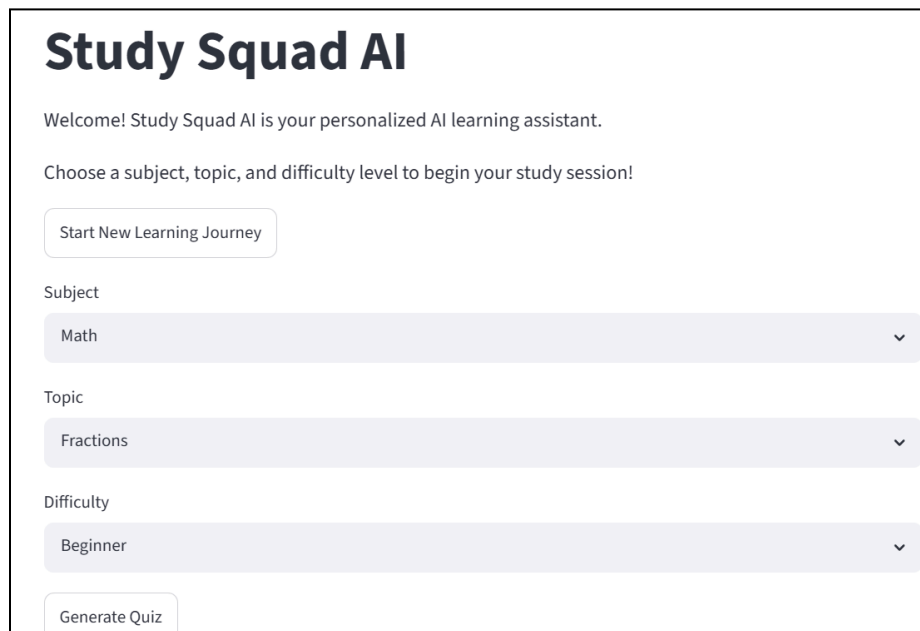
A multi-agent educational learning assistant that generates personalized quizzes, provides AI coaching, recommends curated learning resources, and tracks student progress.

Author: Karthikeyan Anand

Date Submitted: July 5, 2026

GitHub Repository: <https://github.com/k7anand/Study-Squad-AI>

Live App: <https://study-squad-ai-959780117802.us-central1.run.app> [Available until Aug 2026]

The screenshot shows the 'Study Squad AI' web interface. At the top, it says 'Welcome! Study Squad AI is your personalized AI learning assistant.' Below this, it prompts the user to 'Choose a subject, topic, and difficulty level to begin your study session!'. There are three dropdown menus: 'Subject' with 'Math' selected, 'Topic' with 'Fractions' selected, and 'Difficulty' with 'Beginner' selected. A 'Start New Learning Journey' button is located above the dropdowns, and a 'Generate Quiz' button is at the bottom.

Learning is a highly individual process, with every student progressing at their own pace and developing unique strengths and areas of improvement. While assessments can help measure understanding, meaningful learning extends beyond simply answering questions correctly. Students benefit from timely feedback, personalized guidance, and access to relevant learning resources that reinforce concepts and encourage continuous improvement. Study Squad AI was developed for the AI Agents: Intensive Vibe Coding Capstone Project under the track *Agents for Good* with the goal of bringing these elements together into a single intelligent learning assistant that supports students throughout their learning journey, from assessment and feedback to personalized study recommendations and progress tracking.

Delivering this personalized learning experience requires several interconnected tasks to work together seamlessly. Students must first be assessed, their performance evaluated, their learning progress analyzed, and appropriate study resources recommended based on their needs. Rather than relying on a single AI component to perform every task, Study Squad AI adopts a

multi-agent architecture in which specialized agents collaborate to provide these capabilities. A Coordinator Agent orchestrates the overall learning session by delegating responsibilities to a Tutor Agent for assessment generation, a Coach Agent for personalized feedback, a Resource Agent for educational recommendations, and supporting MCP servers for resource retrieval and long-term progress tracking.

Recommended Resources

Hi there! I'm excited to help you start your journey into the world of physics. Here is your quick guide to getting started with **Forces**:

1. Why This Topic is Important

Forces are the foundation of how the physical world works! Understanding forces explains why things move, how they stop, and why objects don't just float away into space. From walking down the street to launching rockets, forces are behind it all.

2. Recommended Study Order

Since we have one excellent introductory resource, you can dive straight into it:

1. **What Is a Force? (The Organic Chemistry Tutor)**

3. Resource Description

- **What Is a Force? (The Organic Chemistry Tutor)**: This video offers a clear, beginner-friendly introduction to the fundamental concept of forces, explaining pushes, pulls, and how they affect objects in an easy-to-follow way.

4. Motivation

You're about to unlock the secrets behind how everything in the universe moves and interacts—you've got this, and happy learning!

What Is a Force?

The Organic Chemistry Tutor

Watch on YouTube

The Tutor Agent is responsible for generating personalized assessments based on the student's selected subject, topic, and difficulty level. Rather than embedding the quiz generation logic directly within the agent, the functionality is implemented as a reusable Agent Skill. By separating quiz generation from the agent itself, the assessment capability becomes modular and can be reused by other agents or applications without duplicating code. This design promotes a clear separation of responsibilities, allowing the Tutor Agent to focus on orchestrating the assessment process while the Agent Skill is responsible for generating high-quality quiz content.

Question 1

What fraction of the circle below is shaded? (Imagine a circle divided into 4 equal parts, with 1 part shaded.)

- ☒ A. $\frac{1}{4}$
☐ B. $\frac{1}{2}$
☐ C. $\frac{3}{4}$
☐ D. $\frac{4}{4}$

While Agent Skills encapsulate reusable AI capabilities, Study Squad AI also leverages Model Context Protocol (MCP) servers to provide external services that can be shared across multiple agents. This separation enables agents to remain focused on reasoning and decision-making while delegating specialized operations to dedicated servers. The Resource MCP Server provides curated educational resources based on a student's selected topic and difficulty level, allowing the Resource Agent to generate personalized study recommendations without maintaining its own resource database. Similarly, the Memory MCP Server manages persistent quiz history by storing and retrieving student performance through a SQLite database. This enables the Coach Agent to analyze learning progress across multiple study sessions and provide feedback based on a student's historical performance rather than a single assessment.

Coach Feedback

Hello! As your academic coach, I am so proud of your dedication to your studies. Let's take a look at where you stand and how we can boost your math scores together!

1. Performance Summary

You are making steady progress! You have a solid foundation started in fractions, and with just a little more focused practice, you can easily turn that into a master strength. Trigonometry is currently challenging you a bit more, which is completely normal—it just means we need to try a different approach to unlock it!

2. Topics to Improve

- **Math | Trigonometry** (Current score: 33.33%)
- **Math | Fractions** (Current score: 66.67%)

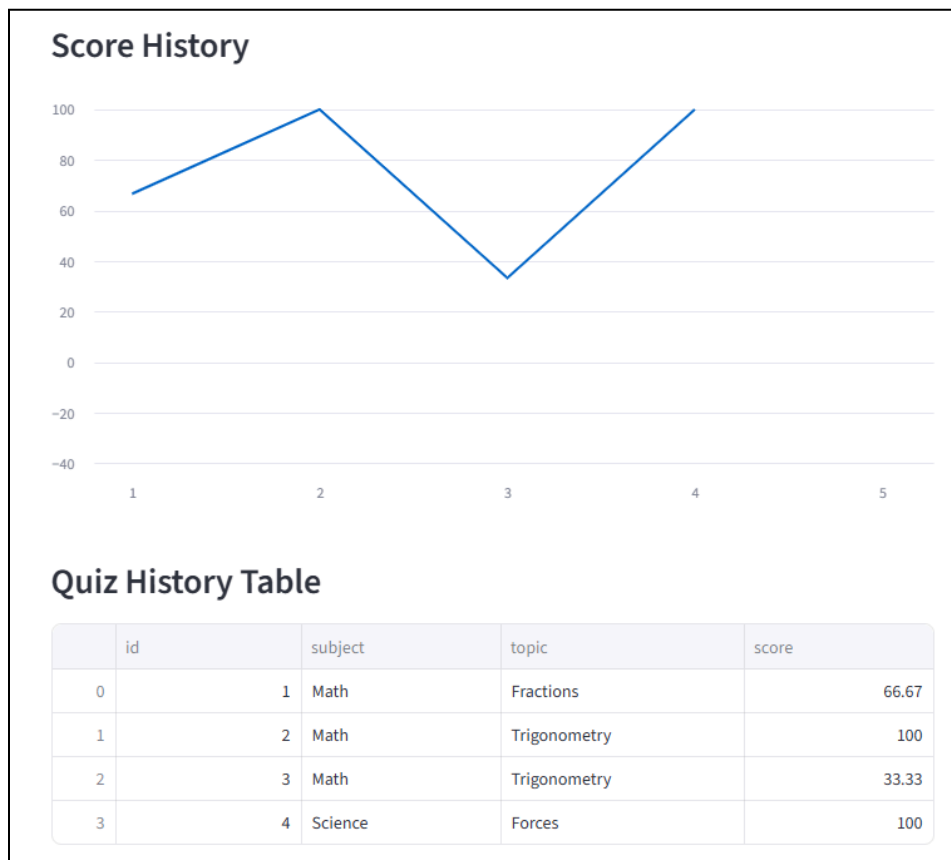
3. Study Recommendations

1. **Master the Triangle Basics (Trigonometry):** Spend 15 minutes a day practicing SOH-CAH-TOA. Draw right-angled triangles, label the Opposite, Adjacent, and Hypotenuse sides, and write out the ratios. Visualizing the triangles is key!
2. **Use Visual Fraction Models (Fractions):** Since you are already close to mastering fractions, practice using fraction circles or bars online to visualize how to find common denominators when adding and subtracting.
3. **The "Teach-Back" Method:** Try explaining a fraction division problem or a basic trig ratio to a friend, family member, or even out loud to yourself. Teaching a concept is one of the fastest ways to solidify it in your own mind!

To support long-term learning, Study Squad AI incorporates a lightweight SQLite database to persist quiz results across study sessions. Following each completed assessment, the Memory

MCP Server records the student's subject, topic, and score, allowing learning progress to be retained even after the application has been restarted. The information has been presented through the Quiz History page of the Streamlit application, where students can review their previous quiz attempts, visualize their performance over time, and identify areas requiring more practice. By combining Python with SQLite, the application provides a simple yet persistent layer that transforms individual assessments into an evolving record of student progress.

The Streamlit application serves as the primary interface through which students interact with Study Squad AI. Through a simple web-based interface, students can select a subject, topic, and difficulty level, complete AI-generated assessments, receive personalized coaching, explore recommended learning resources, and review their historical learning progress. While the underlying system consists of multiple collaborating agents, MCP servers, and a persistent SQLite database, Streamlit presents these capabilities through a unified and interactive user experience, allowing students to focus on learning rather than the underlying implementation.



The reasoning capabilities of the agents are powered by Google's Gemini models. Rather than using a single prompt to perform every task, Study Squad AI assigns Gemini to specialized agents that each focus on a specific responsibility. The Tutor Agent generates personalized multiple-choice assessments, the Coach Agent analyzes quiz performance and produces

individual feedback, and the Resource Agent explains curated educational resources and recommends an effective study order. This specialization enables each agent to use Gemini for a clearly defined objective while the Coordinator Agent orchestrates the overall learning flow.

To improve software quality and simplify development, Study Squad AI includes a dedicated testing suite covering its major architectural components. Individual tests verify the quiz generation skill, agent behavior, MCP servers, database operations, grading functionality, and overall coordination flow. By validating components independently before integrating them into the complete application, the project becomes easier to debug, maintain, and extend while increasing confidence in the correctness of the overall system.

```
(venv) E:\python_projects\study-squad-ai>python -m tests.test_grading

Quiz Results:

Correct: 2
Total: 3
Score: 66.67%

Missed Questions:

{'question_number': 2, 'question': 'Q2', 'correct_answer': 'C', 'student_answer': 'D'}
```

Although the deployed application uses a custom multi-agent architecture tailored to Study Squad AI, this project also includes a demonstration of Google's Agent Development Kit (ADK) to illustrate one of the key concepts covered in the AI Agents: Intensive Vibe Coding course. Specifically, the demonstration shows how the reusable Quiz Generation Skill can be exposed as an ADK FunctionTool, allowing an ADK-based Tutor Agent to access the same assessment generation capability while maintaining the modular design adopted throughout this project.

```
(venv) E:\python_projects\study-squad-ai>python -m tests.test_quiz_tool
QUIZ TOOL CALLED!

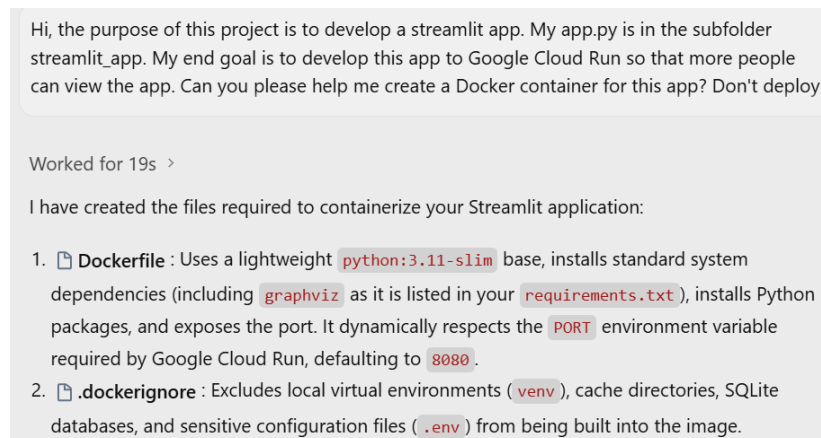
Question 1
Imagine a pie cut into 8 equal slices. If 3 slices are eaten, what fraction of the pie is left?
A. 3/8
B. 5/8
C. 1/8
D. 8/8
Correct Answer: B

Question 2
In the fraction 2/7, what does the number 7 represent?
A. The number of parts being considered.
B. The total number of equal parts in the whole.
C. The size of each part.
D. The number of wholes.
Correct Answer: B

Question 3
Which of the following fractions is equal to 1 whole?
A. 1/4
B. 2/3
C. 5/5
D. 1/2
Correct Answer: C
```

The above figure demonstrates the interaction between the Google ADK and the reusable Quiz Generation Skill. When the test is executed, the ADK Tutor Agent invokes the `quiz_generation_tool`, which then calls the Quiz Generation Skill to generate personalized assessments. The console output confirms this interaction by displaying the message “QUIZ TOOL CALLED!” followed by the generated quiz. This demonstrates the separation of responsibilities within the architecture where the ADK Tutor Agent determines when an assessment should be generated while the Quiz Generation Skill is responsible for how the assessment is created. Exposing the skill as an ADK FunctionTool allows the same functionality to be reused by different agents without duplicating the underlying implementation.

To improve portability and reproducibility of Study Squad AI, the application was containerized using Docker. During development, Google’s Antigravity coding assistant was used to help generate the initial Docker configuration based on the project’s structure and deployment requirements. The generated configuration was subsequently reviewed and integrated into the project to support a consistent execution environment across different machines.



Hi, the purpose of this project is to develop a streamlit app. My app.py is in the subfolder streamlit_app. My end goal is to develop this app to Google Cloud Run so that more people can view the app. Can you please help me create a Docker container for this app? Don't deploy

Worked for 19s >

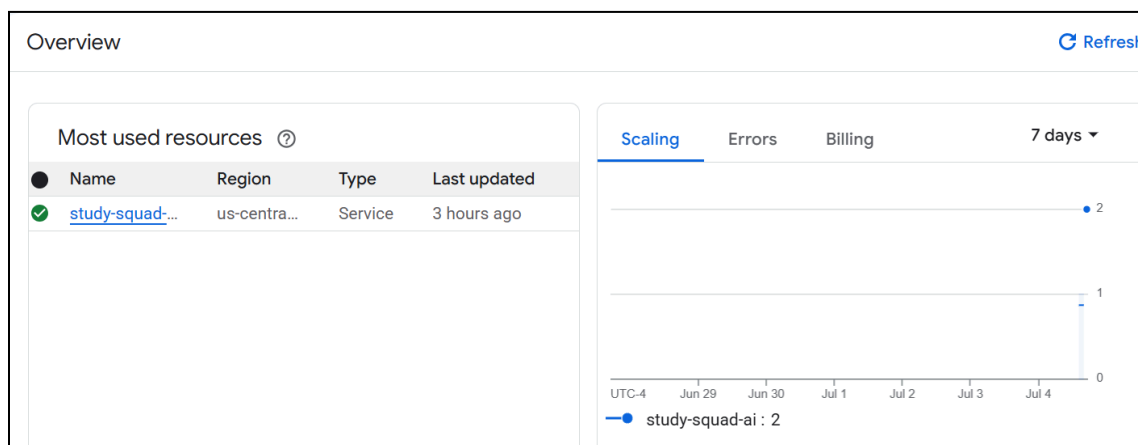
I have created the files required to containerize your Streamlit application:

1. `Dockerfile` : Uses a lightweight `python:3.11-slim` base, installs standard system dependencies (including `graphviz` as it is listed in your `requirements.txt`), installs Python packages, and exposes the port. It dynamically respects the `PORT` environment variable required by Google Cloud Run, defaulting to `8080`.
2. `.dockerignore` : Excludes local virtual environments (`venv`), cache directories, SQLite databases, and sensitive configuration files (`.env`) from being built into the image.

Docker packages the application together with its Python dependencies, system libraries, and runtime configuration into a single container image. As a result, the application can be executed consistently across different operating systems without requiring users to manually install packages or resolve dependency conflicts. The Docker configuration includes a `Dockerfile`, which defines the application’s runtime environment, and `.dockerignore`, which excludes unnecessary configuration files from the container image. This reduces the container size while preventing sensitive information from being included in distribution builds. By containerizing the application, Study Squad AI becomes easier to distribute, reproduce, and deploy, providing a foundation for hosting the application on a managed cloud platform.

After containerizing the application with Docker, Study Squad AI was deployed as a managed cloud service using Google Cloud Run. The Docker image was first built remotely using Google Cloud Build before being deployed to Cloud Run, where the application could be accessed

through a standard web browser without requiring users to download Python or any project dependencies. Environment variables, including the Gemini API key, were securely provided during deployment rather than being embedded with the source code. This deployment approach demonstrates how the application can be packaged, deployed, and made publicly accessible while maintaining a reproducible and portable execution environment.

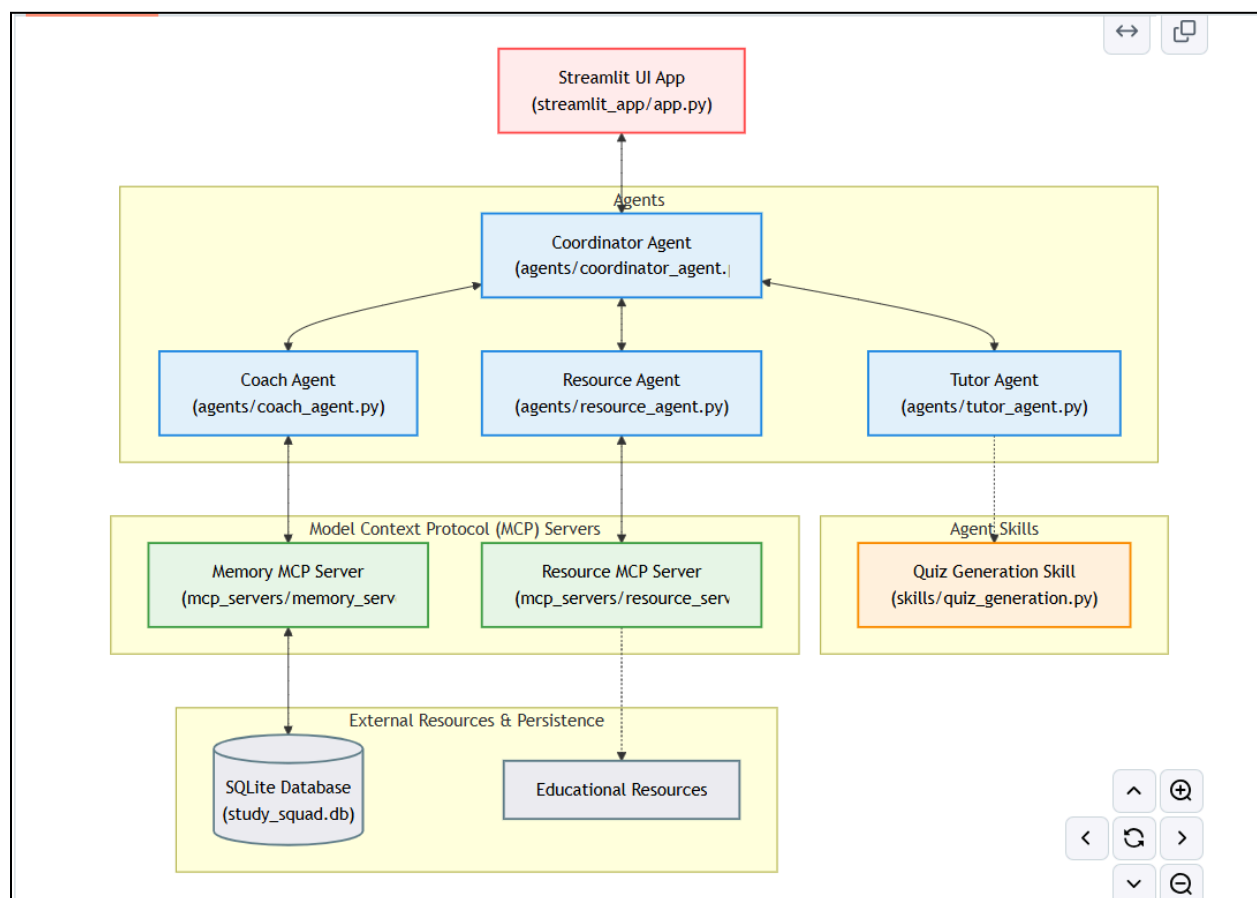


Altogether, the development of Study Squad AI followed a structured software engineering workflow that transformed from a simple idea into a deployable application. The project began with the implementation of a custom multi-agent architecture, where specialized agents were developed to coordinate assessment generation, performance analysis, and personalized resource recommendations through reusable Agent Skills and MCP servers. Each component was independently tested before being integrated into the complete Streamlit application, ensuring that individual functionalities operated correctly prior to system-wide integration. Once the application was fully implemented and validated, it was containerized using Docker and deployed to Cloud Run, enabling users to access the application through a web browser without requiring local installation. This progression, from implementation and testing to packaging and deployment, reflects a practical software development lifecycle while demonstrating how multiple AI technologies can be combined into a cohesive educational application.

Throughout the development of Study Squad AI, I wanted to build an application that I would have found genuinely useful when I was a student and AI agents were not as accessible as they are today. Even though there are countless educational resources available online, identifying high-quality material and deciding what to study next can often be overwhelming. As a result, I envisioned an application that not only generates personalized assessment but also explains areas requiring improvement, recommends relevant learning resources, and encourages students to continue learning. Rather than expecting students to search through numerous videos or websites, the Resource Agent curates appropriate materials based on the selected topic and difficulty level while the Coach Agent provides constructive feedback by considering both the student's current assessment and previous quiz history. Python and SQLite provided a simple and

effective way to implement persistent progress tracking during development, while highlighting how a managed database solution such as Cloud SQL would be a more scalable choice for a production deployment. In addition, Google’s Antigravity proved to be a valuable development assistant by helping generate supporting project files like the Docker configuration, allowing me to focus more on the application’s architecture and functionality. Overall, the project provided an opportunity to combine artificial intelligence, software engineering, cloud deployment, and educational technology into a single application while demonstrating how multiple AI concepts can work together to create a meaningful learning experience.

The following diagram summarizes how the major components of Study Squad AI interact during a learning session, and this diagram is built using Mermaid.js in the LaTeX editor.



Although Study Squad AI demonstrates a complete end-to-end intelligent learning workflow, several opportunities exist for future development. The current implementation uses SQLite to persist quiz history, which provides a lightweight solution for local development and demonstration purposes. For larger-scale deployments supporting multiple concurrent users, a managed relational database such as Cloud SQL would offer greater scalability, reliability, and data management capabilities. Additional enhancements could include user authentication to

provide intellectualized learning profiles, expanding the range of supported academic subjects beyond Math and Science, incorporating adaptive learning pathways that automatically adjust assessment difficulty based on student performance, and developing richer learning analytics to help students visualize their long-term progress. Also, the Resource MCP Server could be extended to integrate additional educational providers beyond the current curated resource collection, enabling students to access a broader range of high-quality learning materials. These enhancements would further strengthen Study Squad AI as a personalized educational platform while preserving the modular multi-agent architecture developed in this project.